

Beam Search Algorithm in Artificial Intelligence

- Beam Search only keeps the best (lowest-h) n nodes on open list
- n is the “beam width”
 - $n = 1$, Hill climbing
 - $n = \text{infinity}$, Best first search

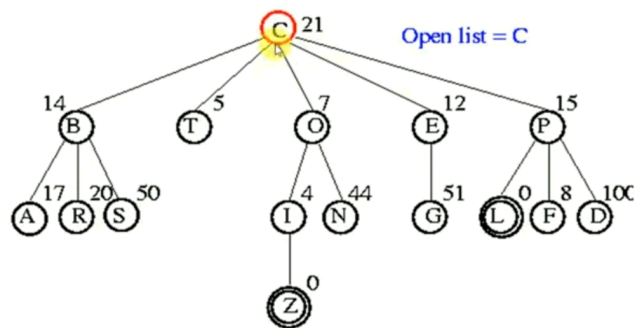
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the current node and everything will be placed into

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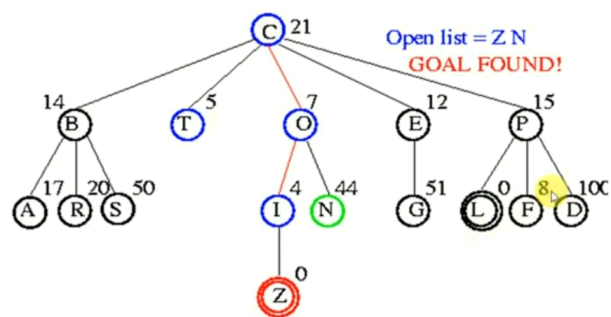
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open list first
now the successors

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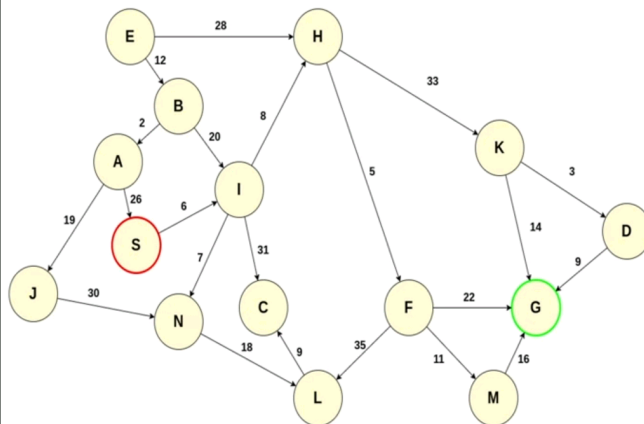
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comparisons in this case
if you have a hilly

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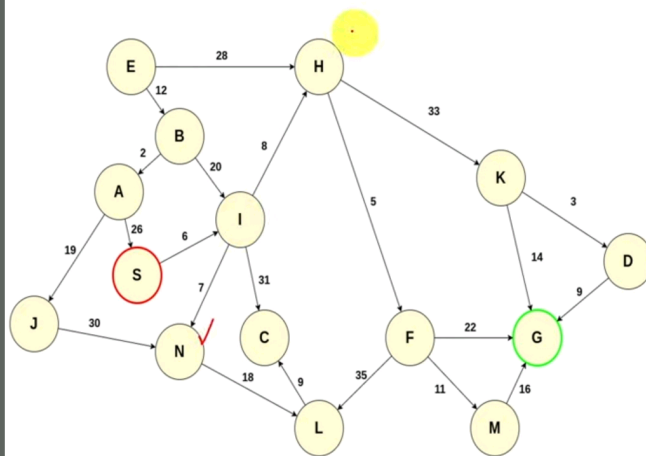
- Initialize the tree and add **S** to the **closed list**:



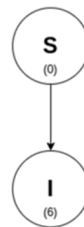
Node[cost]
I[6]

Closed List
S

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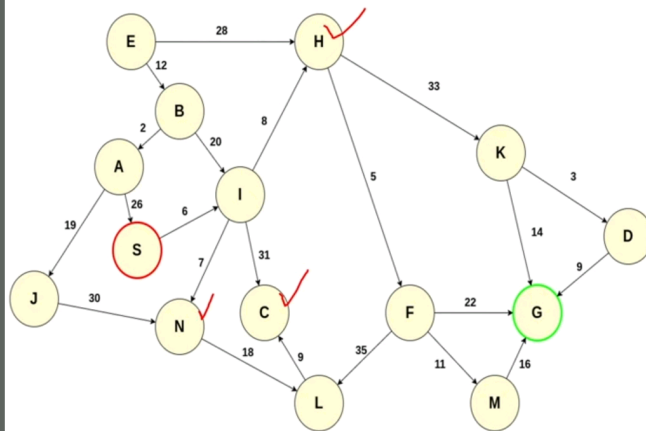
- S only has one adjacent node, so it is added to the tree:



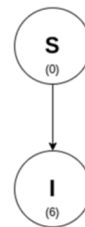
Node[cost]
N[13]
H[14]
C[37]

Closed List
S
I

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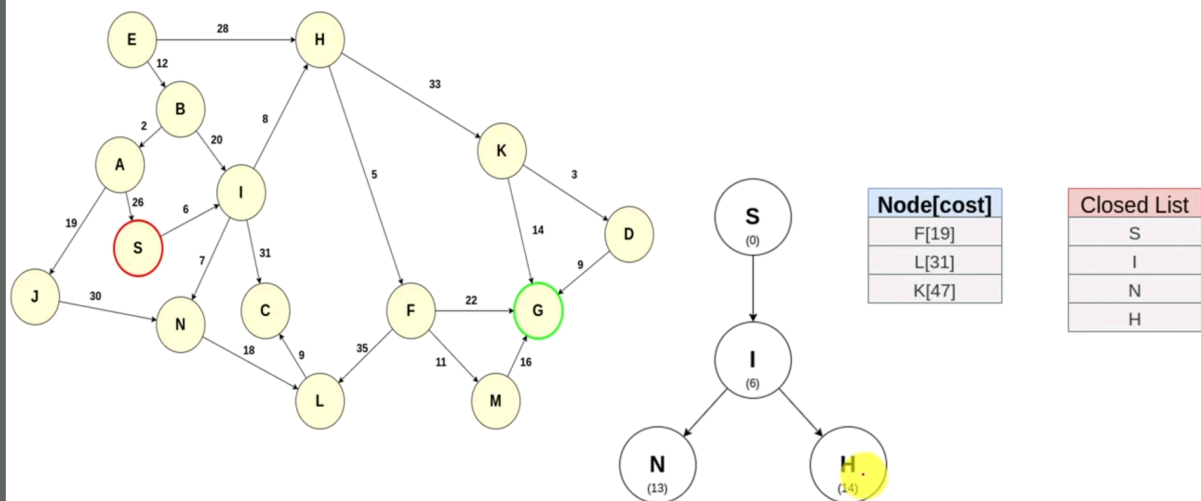
- S only has one adjacent node, so it is added to the tree:



Node[cost]
N[13] ✓
H[14] ✓
C[37] ✓

Closed List
S
I

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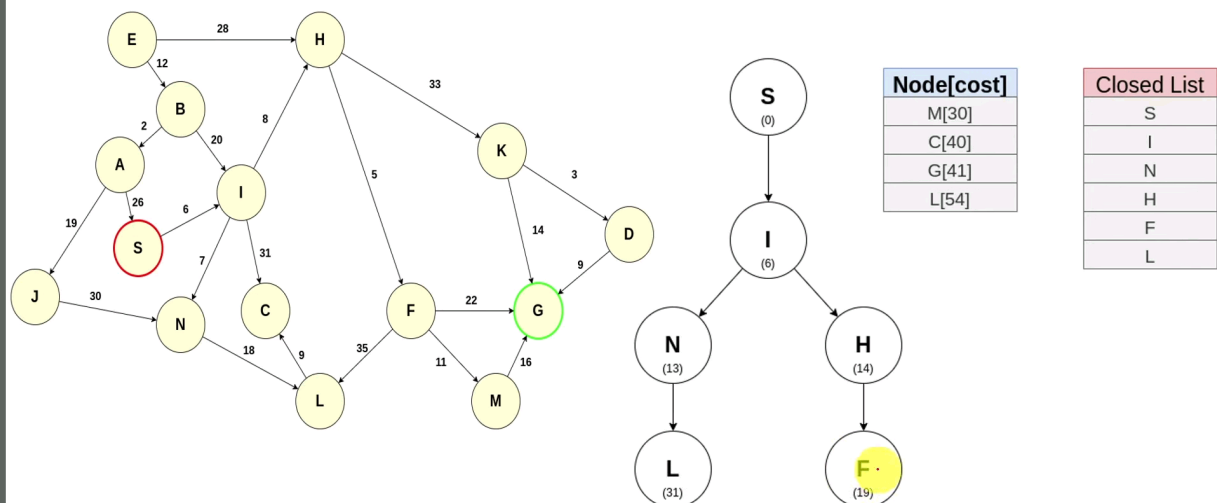


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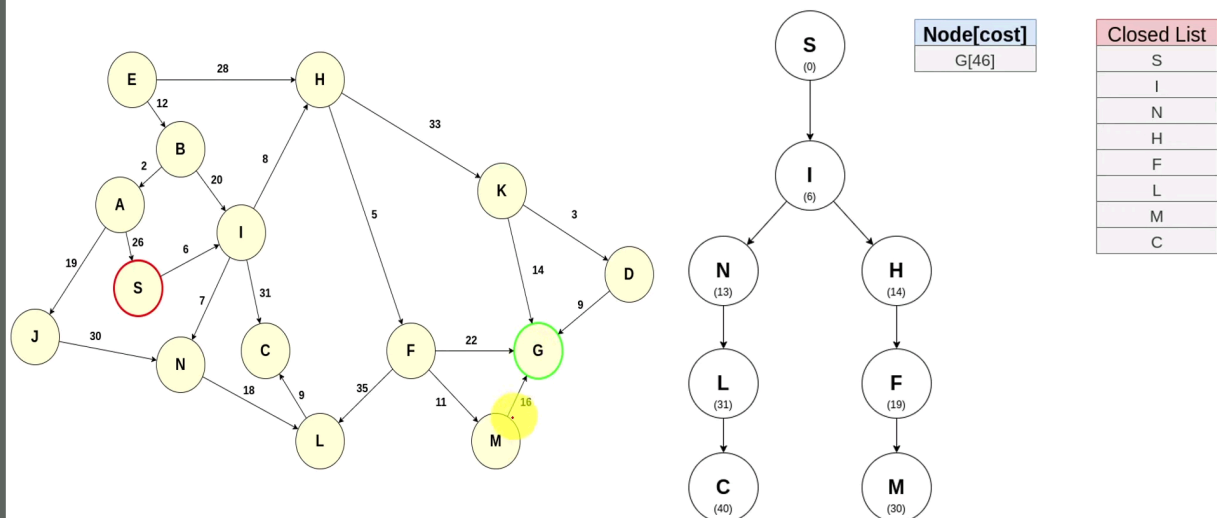


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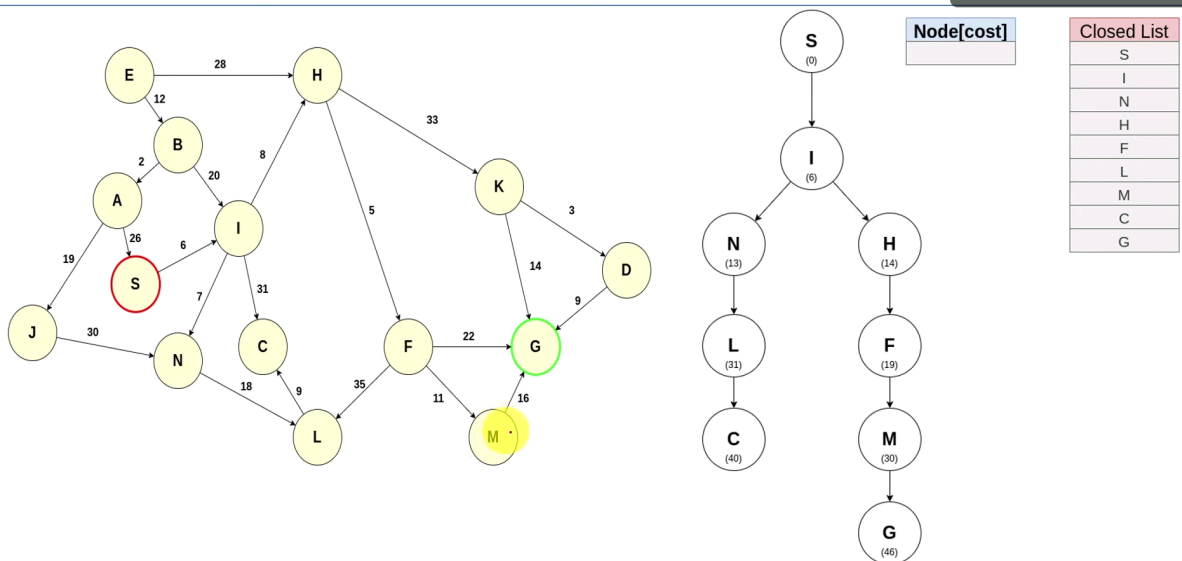
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3. Principal Component

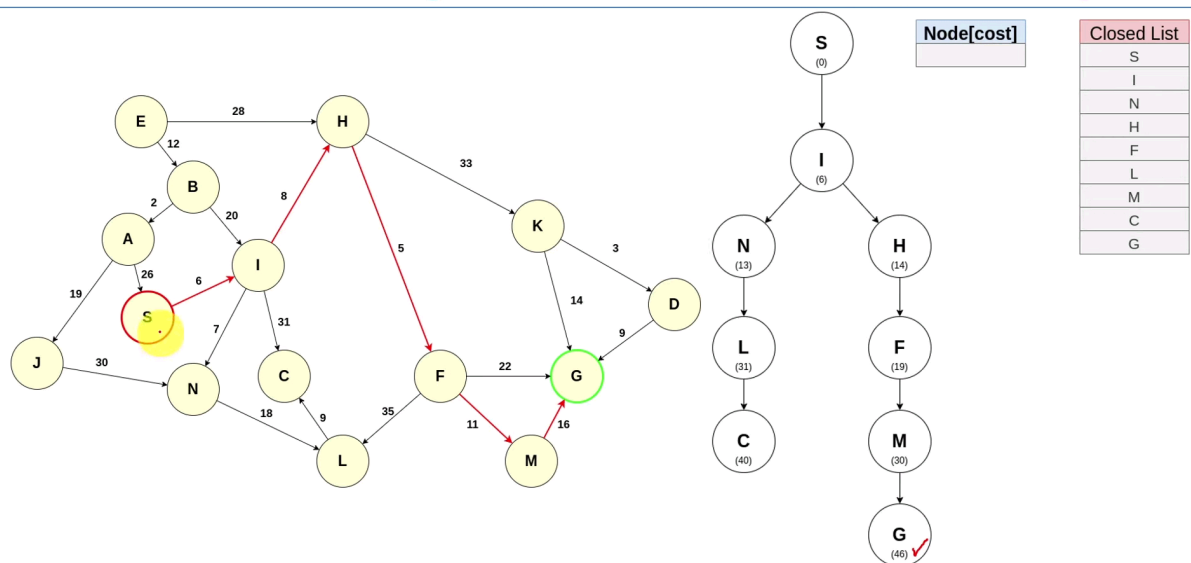


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